MoveOn Phone Cover Downloader & Translation Pipeline

Specification Version: 1.1 | **Project:** inventory/coverstore | **Status:** Production

1. System Architecture & Constraints

Automates single-product asset acquisition from **moveon.global**. Strictly executes against only the user-provided product URL with zero out-of-scope spidering. Auto-identifies phone cover materials, prompts for phone model selection, generates standardized SKU codes, and replaces non-English text on photos with clean English translations.

2. Custom SKU Generation Standard

All variation cover images are named using the standardized format:
[model abbreviation]-[material abbreviation]-[color abbreviation]-[print abbreviation]
Example: RN8-SIL-WT-S.jpg (Realme Neo8, Liquid Silicone, Antique White, Solid Color)

Component	Code Format	Representative Examples
Phone Model	2-6 Alphanumeric	`RN8 (Realme Neo8), IP15PM (iPhone 15 Pro Max), S24U (Galaxy S24 Ultra)`
Material	3 Letters	`SIL (Liquid Silicone), GLS (Glass), TPU (TPU), LTH (Leather), PC (Hard PC)`
Color	2 Letters	`WT (White), BK (Black), PR (Purple), GN (Green), BL (Blue), CL (Clear)`
Print / Finish	1 Letter	`S (Solid/Cube), P (Printed), G (Gradient), E (Electroplated/Mirror), C (Clear)`

3. Storage Hierarchy

Directory Path	File Naming & Contents
`\[model]\[material]\original\`	original_01.jpg ... Product showcase & gallery images.
`\[model]\[material]\sku\`	[SKU].jpg (e.g. RN8-SIL-WT-S.jpg) ... Variant covers.
`\[model]\[material]\translated\`	translated_[SKU].jpg ... English-translated graphics.
`\scripts\`	downloader.py, sku_generator.py, moveon_client.py, translate_helper.py.